

Cross-Lingual Transfer Learning for Low-Resource Sentiment Analysis

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Abstract

We address the challenge of sentiment analysis in low-resource languages where labeled training data is scarce. Our approach, CrossSent, leverages multilingual pretrained language models with a novel alignment-then-transfer framework: we first align sentiment-bearing representations across languages using parallel corpora, then fine-tune on high-resource English data and transfer to 12 target languages. CrossSent achieves an average F1-score of 82.4% across target languages, outperforming zero-shot multilingual BERT by 6.8 percentage points and few-shot approaches using 100 labeled examples by 3.1 points. Analysis reveals that alignment quality correlates strongly with typological similarity ($r=0.73$), suggesting that morphologically rich languages benefit most from explicit representation alignment.

1. Introduction

Sentiment analysis has achieved remarkable accuracy in high-resource languages such as English and Chinese, driven by large annotated datasets and powerful pretrained models [1, 2]. However, extending these advances to the vast majority of the world’s 7,000+ languages remains a significant challenge. For most languages, labeled sentiment data is either nonexistent or limited to a few hundred examples, making supervised fine-tuning unreliable.

Cross-lingual transfer learning offers a promising path forward. Multilingual pretrained models such as mBERT [3] and XLM-R [4] learn shared representations across languages during pretraining, enabling zero-shot transfer: a model fine-tuned on English can be applied directly to other languages. However, zero-shot performance degrades significantly for languages distant from English, particularly for sentiment-specific nuances that may not align across cultural and linguistic boundaries.

We propose CrossSent, a two-stage framework that explicitly aligns sentiment representations before transfer. In the alignment stage, we use parallel sentence pairs to learn a projection that maps sentiment-bearing features into a shared space. In the transfer stage, we fine-tune on English sentiment data in this aligned space and apply the model to target languages without additional labeled data. This decoupling allows each stage to be optimized independently and provides interpretable diagnostics of transfer quality.

2. Method

CrossSent builds on XLM-R-Base (270M parameters) as the backbone encoder. The alignment stage uses 50K parallel sentence pairs from OPUS [5] for each language pair. We add a lightweight projection head (2-layer MLP, hidden size 256) on top of the [CLS] representation and train with a contrastive loss that pulls parallel sentences together while pushing non-parallel pairs apart in the sentiment-relevant subspace.

The transfer stage fine-tunes the aligned encoder on the SST-2 training set (67K English examples) with a standard cross-entropy loss. At inference time, target-language inputs are encoded by the same aligned encoder and classified by the English-trained head. We optionally apply a temperature-scaled calibration step using 20 unlabeled target-language examples to correct for distribution shift in the output logits.

3. Experiments

We evaluate on the multilingual Amazon Reviews dataset [6] covering 12 languages: French, German, Spanish, Japanese, Korean, Arabic, Hindi, Thai, Vietnamese, Turkish, Swahili, and Yoruba. We report binary sentiment classification F1-scores on the standard test splits. Baselines include zero-shot mBERT, zero-shot XLM-R, and few-shot XLM-R with 100 target-language labeled examples. Table 1 summarizes the main results.

Table 1. F1-score (%) by language resource level (High / Mid / Low) and overall average.

Method	High	Mid	Low	Avg
mBERT (zero)	78.3	72.1	64.5	71.6
XLM-R (zero)	81.5	76.8	69.2	75.8
XLM-R (100-shot)	84.1	79.6	73.8	79.2
CrossSent (ours)	87.2	83.5	76.4	82.4

4. Conclusion

We presented CrossSent, a cross-lingual transfer framework that achieves strong sentiment analysis performance in low-resource languages through explicit representation alignment. Our analysis highlights the importance of typological similarity in transfer success and suggests that targeted alignment is most beneficial for morphologically divergent language pairs. Future work will explore extending CrossSent to aspect-based sentiment analysis and investigating alignment strategies for code-switched text.

References

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